

Silent Support System

A private, AI-assisted pathway to human support for students who may be struggling but do not know how to ask for help.

Problem · Solution · What changed · Privacy · Testing · Setup

AT A GLANCE

What this document covers

This document explains the **Silent Support System**, a student-wellbeing feature added to the existing 21-page NeX Camp website. It describes the problem it addresses, how the solution works for students and for the support team, exactly what was changed in the project, and how privacy and safety are protected.

Item	Summary
Feature	Silent Support System — optional wellbeing check-in, “I don't know how to ask for help”, private/anonymous support requests, “check on me later”, support-team workflow and follow-up.
Pages used	Existing page 02 Student Dashboard and existing page 10 Mission Control. No new page, route or navigation item was created — the site still has exactly 21 pages.
AI role	Rules decide the support level; AI may only re-word the supportive message and is blocked from clinical language. Works fully without AI.
Human role	A support person (new COUNSELLOR role) is assigned, contacts the student, schedules follow-up and records the outcome.
Not	Not a medical or diagnostic system. It never says a student has any condition.
Status	Implemented, tested (24 new automated tests passing; 0 existing tests broken), committed to branch claudergracious-ritchie-wtfukt.

SECTION 1

The problem

Many students who are struggling — with stress, studies, loneliness or something more serious — never ask for help. Not because help does not exist, but because asking is hard. The existing website already helped students with attendance, mess, hostel, complaints and gate passes, but it had **no private way** for a student to signal that they were not doing well.

Why students stay silent

- **They do not know how to say it.** Writing “I need help” to a staff member feels like a big, formal step.
- **They worry about who will see it.** Teachers, wardens or other students finding out is a real fear.
- **They are not ready to talk yet,** but would welcome someone checking on them later.
- **Early signs go unnoticed.** Without a gentle, regular check-in, difficulty is only seen when it becomes a crisis.
- **When help is requested, there is no follow-through.** Without a tracked workflow, a request can be lost, and nobody records whether support actually happened.

Risks a careless solution would create

Risk	Why it matters
Labelling or diagnosing students	An app saying “you have depression” is harmful, inaccurate and not its role.
Exposing private feelings	Wellbeing answers seen by teachers, admins, other students or public dashboards would destroy trust.
AI making decisions alone	Safety decisions must stay with qualified people, not a model.
Invented emergency numbers	A wrong hard-coded helpline is dangerous in a real crisis.
Breaking the existing website	The 21 existing pages, APIs and workflows must keep working.

Goal: make asking for help effortless and private, notice early when a student may benefit from support, hand them to a real person, and make sure someone follows up — without diagnosing anyone and without creating a single new page.

SECTION 2

The solution

The Silent Support System adds a small, optional and private support layer to two pages that already exist. The AI assists; people decide and act.

End-to-end flow



Problem	Solution in the website
Students do not know how to ask	A prominent "I don't know how to ask for help" button with one-tap options: counsellor, mentor/faculty, private conversation, anonymous, check on me later, emergency resources. No writing needed.
Early signs go unnoticed	An optional 30-second wellbeing check-in ("How are you feeling today?") with four emoji-scale questions. Every question and the whole check-in can be skipped.
Fear of being exposed	Only the student and the support team can see anything. Anonymous requests hide the student's name and ID. Admins see only totals.
Not ready to talk	"I don't want to talk right now. Check on me later." — pick 1 day, 3 days or 1 week; the system reminds both the student and the team.
No follow-through	A tracked workflow: Request created → Support team assigned → Contacted → Follow-up → Resolved , with the outcome recorded. The student sees progress.
Crisis situations	"I don't feel safe right now" shows a calm safety panel with the institution's own emergency contacts and alerts the support team immediately.
Diagnosis risk	Support-routing levels only, in plain supportive language. AI wording containing clinical terms is automatically rejected.

Support levels (routing only — not diagnoses)

Level	When	What the student is told
Stable	Answers are mostly okay	"Thanks for checking in... the support team is here whenever you need them."
Could benefit from support	Some difficulty reported	"You may benefit from a short conversation with someone — it's completely your choice."
Support recommended	Strong difficulty, or the same difficulty in more than one recent check-in	"Your recent check-ins suggest that additional support may be helpful. Would you like to talk to someone privately?"
Immediate human attention required	Student taps "I don't feel safe right now"	A safety panel: "Please reach a person right now", configured emergency contacts, and the support team is alerted.

SECTION 3

What the student sees — page 02 Student Dashboard

The card appears on the existing Student Dashboard, below the existing summary, in the website's own dark theme, typography and buttons. It is visible only to signed-in students.

YOUR WELLBEING MATTERS · PRIVATE Only you and the student support team can see this.

How are you feeling today?

YOUR PRIVATE REQUEST · SUP-000001 Talk to a mentor / faculty member

Request created Support team assigned Contacted Follow-up Resolved

WITHDRAW REQUEST

• **SUPPORT RECOMMENDED** Support suggestion — not a diagnosis

Your recent check-ins suggest that additional support may be helpful. Would you like to talk to someone privately?

- You've been feeling low recently.
- Studying has felt difficult recently.
- Asking for help feels hard right now — that's common, and there are quiet ways to do it.
- Your recent check-ins suggest things have felt hard for a while, not just today.

Next: You can request a private conversation — no explanation needed.

WORDING · RULE-BASED · support level decided by fixed rules · a person always makes the decisions

CHECK IN AGAIN

I don't know how to ask for help **EMERGENCY & HELP RESOURCES**

Pick whatever feels easiest. You won't need to write anything.

Ask anonymously

Your name and ID stay hidden.
Replies arrive in your inbox.

I don't want to talk right now. Check on me later.

Nothing starts now. We'll simply check in later.

Talk to a mentor or faculty member

Someone who can help with studies and workload.

Talk to a counsellor

A trained counsellor reaches out privately.

Request a private support conversation

No explanation needed — just a conversation.

View emergency & help resources

Contacts set by your institution.

This is a support tool, not a medical assessment. Only you and the student support team can see this. Your answers are never shown to teachers, wardens or administrators.

After a check-in: a gentle support suggestion, the reasons in plain language, and the “I don't know how to ask for help” options.

YOUR WELLBEING MATTERS · PRIVATE

Only you and the student support team can see this.

How are you feeling today?

YOUR PRIVATE REQUEST · SUP-000002

Talk to a mentor / faculty member

Request created

Support team assigned

Contacted

Follow-up

Resolved

WITHDRAW REQUEST

SUPPORT RECOMMENDED

Support suggestion — not a diagnosis

Your recent check-ins suggest that additional support may be helpful. Would you like to talk to someone privately?

- You've been feeling low recently.
- Studying has felt difficult recently.
- Asking for help feels hard right now — that's common, and there are quiet ways to do it.
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CHECK IN AGAIN

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EMERGENCY & HELP RESOURCES

Pick whatever feels easiest. You won't need to write anything.

Ask anonymously

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Nothing starts now. We'll simply check in later.

Talk to a mentor or faculty member

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Talk to a counsellor

A trained counsellor reaches out privately.

Request a private support conversation

No explanation needed — just a conversation.

View emergency & help resources

Contacts set by your institution.

Your private support request has been received. Someone will reach out — you don't need to explain anything.

This is a support tool, not a medical assessment. Only you and the student support team can see this. Your answers are never shown to teachers, wardens or administrators.

After “Request private support”: the request tracker (Request created → Support team assigned → Contacted → Follow-up → Resolved) and a confirmation.

Safety screen on a phone

When a student taps “I don't feel safe right now”, the system does not try to handle the situation itself. It shows a calm message, the emergency contacts configured by the institution (or generic guidance if none are configured — no invented number), and alerts the support team. The layout fits a 390 px phone screen with no sideways scrolling.

YOUR WELLBEING MATTERS · PRIVATE

How are you feeling today?

Only you and the student support team can see this.

YOUR PRIVATE REQUEST · SUP-000002

Talk to a mentor / faculty member

- Request created
- Support team assigned
- Contacted
- Follow-up
- Resolved

WITHDRAW REQUEST

Support suggestion — not a diagnosis

Thank you for telling us. You don't have to handle this alone — please reach a person right now using the contacts below.

Next: Contact someone now using the contacts below. The student support team can also reach out to you.

WORDING · RULE-BASED

· support level decided by fixed rules · a person always makes the decisions

CHECK IN AGAIN

Please reach a person right now.

You don't have to handle this alone. Use one of the contacts below, or tell any staff member you trust.

Your institution has not added its emergency contacts here yet. If you are in immediate danger, call your local emergency services or go to the nearest hospital, and tell your warden or any staff member you trust right now.

I don't know how to ask for help

EMERGENCY & HELP RESOURCES

What happens

- No diagnosis and no alarming label is shown to the student.
- The safety message is fixed text — AI is never used on this path.
- Emergency contacts come from the institution's configuration.
- The support team gets an **urgent** request at once (can be switched off with `SUPPORT_AUTO_ESCALATE_IMMEDIATE=false`).
- The help options stay available below the safety panel.

Student notifications (existing inbox)

- “Your private support request has been received.”
- “A support person has been assigned to your request.”
- “The support team has reached out.” (with an optional short message)
- “Checking in, as you asked.” (when a follow-up is due)

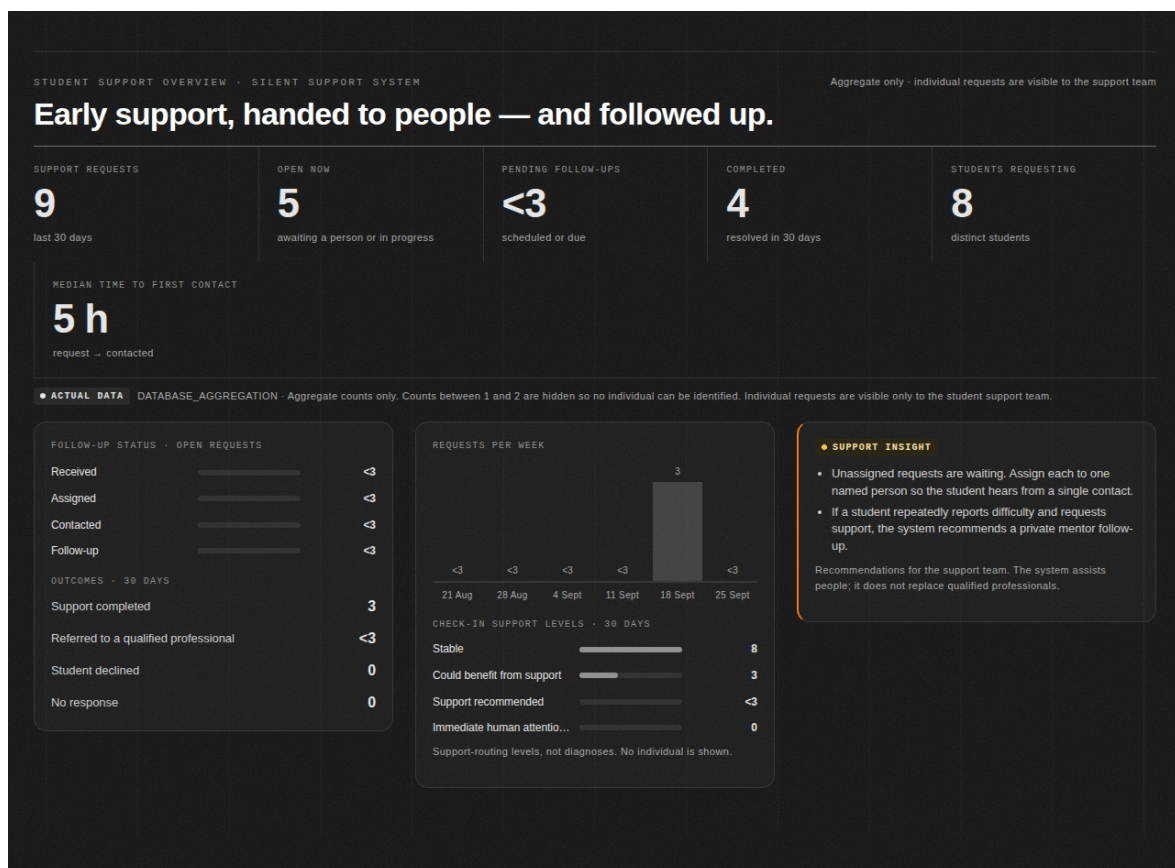
SECTION 4

Support team and admin — page 10

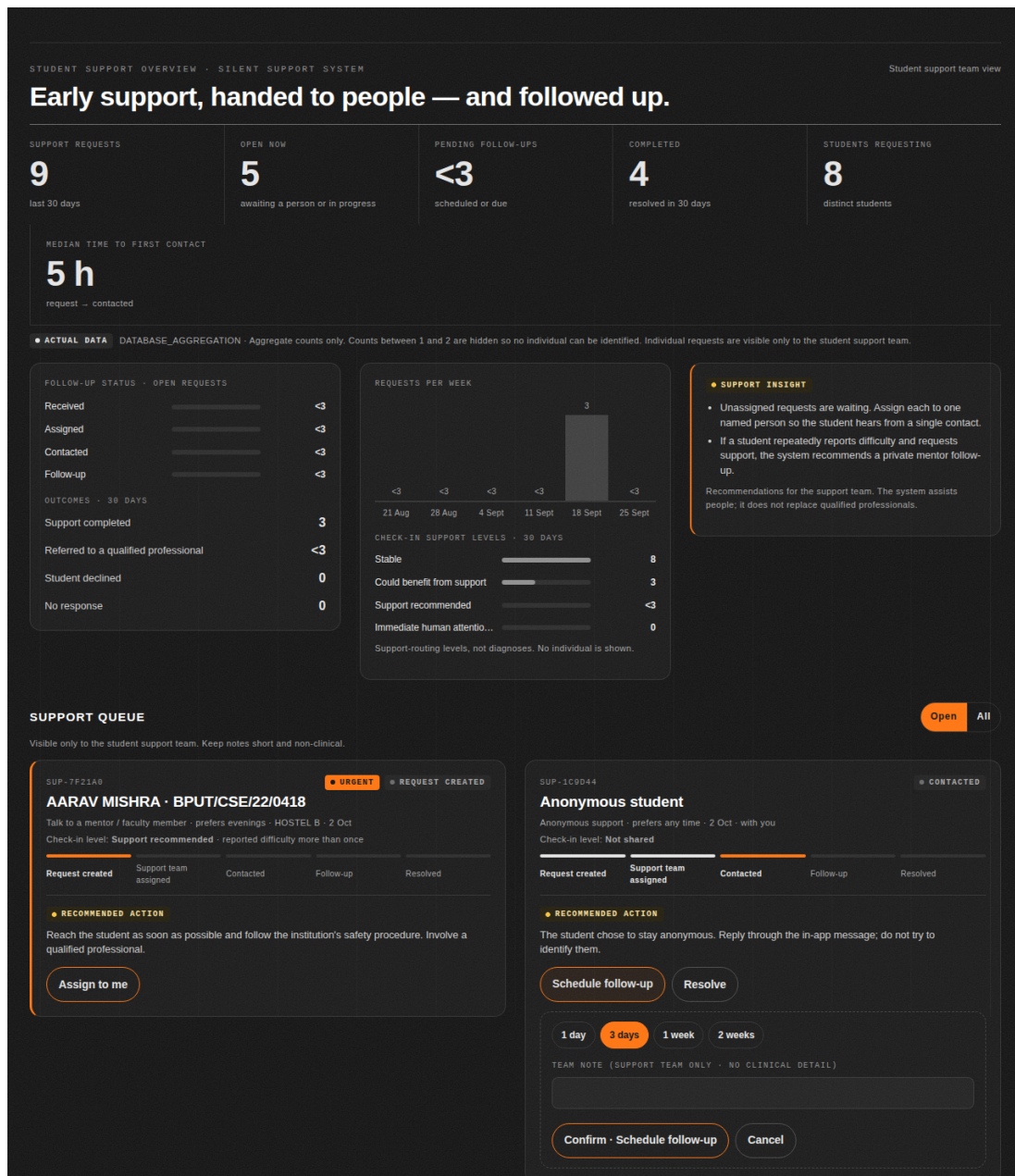
Mission Control

Two panels were added to the existing Mission Control page. What appears depends on the signed-in role.

Role	Sees	Cannot see
COUNSELLOR (student support team)	Overview with exact counts, and the support queue: student name and ID (unless anonymous), the kind of help asked for, preferred time, shared check-in level, recommended action, workflow buttons.	Check-in answers (never shared with anyone).
ADMIN	Student Support Overview with totals only. Any count of 1–2 is shown as “<3” so no individual can be identified.	Any individual request, name or reference.
Warden, facility, mess staff, teachers, other students	Nothing from this feature.	Everything — the API refuses them.



Admin view: Student Support Overview — support requests, open, pending follow-ups, completed, students requesting, time to first contact, follow-up status, weekly requests, check-in levels and Support Insight. Small counts are hidden.



Support team view: the queue. Each case shows its progress, a RECOMMENDED ACTION, and the next step (Assign to me → Mark contacted → Schedule follow-up → Resolve with an outcome). Anonymous cases never show who the student is.

SECTION 5

How the AI is used — and limited

The project already had an AI layer (a provider client that works with Anthropic or any OpenAI-compatible service, with a rule-based fallback). The Silent Support System reuses it rather than adding a new AI service or a new API key.

Step	Who does it	Detail
Decide the support level	Fixed rules (supportRules.js)	Four answers on a 1–5 scale give points of concern. Repeated difficulty within 21 days raises the level. “Not safe” always means immediate attention. A model can never change the level.
Write the supportive message	AI (optional) or rules	The model receives only the level and plain reason lines — never the student's name, ID or answers. Limited to 6 seconds.
Check the AI wording	Language guard	Any output containing clinical words (depression, disorder, diagnosis, clinical, psychiatric...) is rejected and the rule-based text is used instead.
Safety message	Fixed text	AI is never called when a student says they do not feel safe.
Recommend an action to staff	Rules	e.g. “Consider a private mentor check-in focused on workload” — a recommendation for a person, not a prediction about the student.
Decide and act	Humans	Assigning, contacting, follow-up and the outcome are always done by the support team.

Graceful fallback

- **No AI configured:** the feature works completely using rule-based wording (labelled `AI_NOT_CONFIGURED`).
- **AI provider down or slow:** rule-based wording is used (labelled `DETERMINISTIC_FALLBACK`).
- **Support service unreachable:** the card shows “Support check-in is temporarily unavailable. You can still request human support.”

Safe language by design. The system says things like “You may benefit from additional support”, “Would you like to talk to someone?” and “Your recent check-ins suggest that additional support may be helpful.” It never says a student has depression, a mental illness or any disorder. Every message shown to students is also labelled “Support suggestion — not a diagnosis”.

SECTION 6

Privacy and security

Measure	How it is done
Narrow new role	New COUNSELLOR role, deliberately not added to the general staff roles, so it gains no other campus access, and no existing staff role (admin included) can open an individual case.
Minimum data	No free text from students. A check-in stores four optional numbers, a safety flag and the level. Requests store the level only if the student ticks “share my latest check-in level” — answers are never shared.
Automatic deletion	Check-ins expire automatically after 90 days (configurable).
Anonymous option	The student's name and ID are never sent to the support team; replies arrive in the student's own inbox.
Aggregate-only admin view	Totals only; counts of 1-2 are hidden to prevent identifying a person.
Generic notifications	Notification text never contains answers, levels or names.
Existing leak fixed	The help-desk kiosk (page 12) showed a looked-up student's recent notifications to the operator. Support notifications are now excluded there.
Nothing stored in the browser	No offline cache or queue is used for this feature; server responses are marked no-store.
No sensitive data in URLs or logs	Data travels in request bodies; URLs only carry an opaque id. Answers are never logged.
Abuse limits	Rate limit on requests (30 per 10 minutes) and at most 8 check-ins per day.
Existing authentication	Uses the project's existing JWT login and role checks — no separate security system.

SECTION 7

What was changed in the project

The guiding rule was **“add, don't destroy”**. Every change to an existing file is a small insertion marked `SUPPORT HOOK`. Nothing was removed, renamed or redesigned.

New files

Area	Files	Purpose
Backend models	models/support/SupportCase.js, WellbeingCheckIn.js	Two new database collections: support requests and check-ins.
Backend logic	services/support/supportRules.js, supportAi.js, supportConfig.js, supportService.js	Support levels, AI wording with safety filter, configured contacts, workflow, notifications, follow-ups, overview.
Backend API	controllers/support/supportController.js, routes/support/index.js	The /api/support endpoints with role checks and validation.
Demo data	seed/support/index.js, run.js	Support-team account and a few weeks of sample history.
Frontend	src/support/StudentSupport.jsx, MissionSupport.jsx, SupportEntry.jsx, api.js, support.css	Student card, Mission Control panels, lazy loading, API client, styles.
Tests	test/support.test.js, test/support-integration.test.js	24 unit/route tests; 6 database tests.
Docs	CHANGES-SILENT-SUPPORT.md, this PDF	Change record and documentation.

Existing files touched (small insertions only)

File	Change
backend/src/config/constants.js	Added COUNSELLOR role (outside staff roles), support vocabularies, 4 notification types.
backend/src/services/notificationPriority.js	Priority grading for the 4 new notification types.
backend/src/controllers/kioskController.js	Kiosk no longer shows support notifications to operators.
backend/src/routes/index.js	Mounts the new /support router after all existing routers.
backend/src/services/ext/extMonitor.js	“Check on me later” reminders use the existing 60-second timer.
backend/src/seed/index.js, package.json	Support demo data in the seed; new seed:support script.
backend/.env.example	New optional <code>SUPPORT_*</code> settings.
frontend/src/NexCamp.jsx	One import, one slot on page 02, one slot on page 10, support account in the sign-in hint.
README.md	Short feature section and demo account.

New API endpoints (all under /api/support, sign-in required)

Endpoint	Who	Purpose
GET /me	Student	Own latest level, requests and resources
POST /check-in	Student	Submit the optional check-in
POST /requests	Student	Request private support (counsellor, mentor, conversation, anonymous)
POST /check-later	Student	“Check on me later” in 1, 3 or 7 days
POST /requests/:id/withdraw	Student (owner)	Withdraw own request
GET /resources	Anyone signed in	Institution-configured help and emergency contacts
GET /queue	Counsellor	Support queue
POST /cases/:id/status	Counsellor	Assign, contact, follow-up, resolve with outcome
GET /overview	Counsellor, Admin	Aggregate statistics and insights

SECTION 8

Testing and verification

Check	Result
Backend automated tests	411 passed, 0 failed (387 existing + 24 new). 60 skipped because they need a MongoDB database, which was not available in the build environment (54 existing + 6 new).
Frontend production build	Clean. The feature loads as separate small files only when needed, so other pages are not slowed down.
All 21 pages	The repository's own page-check script: 42 of 42 page visits clean (every page, desktop and 390 px phone in low-bandwidth mode).
New screens in a browser	Student, safety, admin and counsellor views exercised end to end: no console errors, no sideways scrolling on phones.
Page count	Unchanged: 21 pages, no new route or navigation item.

What the new tests prove

- Each support level is chosen correctly, repeated difficulty raises it, and “not safe” always gives immediate attention.
- Every rule-based message passes the non-clinical language check; clinical AI wording is rejected.
- AI wording is used only when safe and is clearly labelled; provider failure falls back gracefully; AI is never called on the safety path.
- Emergency contacts come only from configuration — none are invented.
- The student never sees staff notes; the support team never sees answers or phone numbers; anonymous requests hide identity.
- Only COUNSELLOR reaches the queue; ADMIN reaches only the overview; all other roles are refused; every route rejects signed-out users.
- Database tests (run where MongoDB is available) cover the whole workflow, notifications, follow-up reminders, urgent escalation and admin count hiding.

Honest note: the 6 database tests could not run in the build environment because MongoDB could not be downloaded there. They are written to run automatically with npm test wherever MongoDB is available, as the project's other integration tests do.

SECTION 9

Setup, demo and limitations

Configuration (backend/.env)

```
SUPPORT_EMERGENCY_CONTACTS=Campus Security (24x7)|<number>;Medical Centre|<extension>
SUPPORT_HELP_RESOURCES=Counselling Centre|Room <n>, Admin Block|Mon–Sat 9:00–17:00
SUPPORT_AUTO_ESCALATE_IMMEDIATE=true
SUPPORT_CHECKIN_RETENTION_DAYS=90
SUPPORT_MIN_CELL_SIZE=3
```

All settings are optional. AI wording uses the existing AI_PROVIDER, AI_API_KEY and AI_MODEL settings.

Two-minute demo

```
cd backend && npm run seed -- --fresh && npm run dev
cd frontend && npm run dev
```

#	Account	Steps
1	Student — pritish@bput.ac.in / Campus@2026 (signed in automatically)	Page 02: answer the check-in with low “feeling” and “studying” → Could benefit from support. Check in the same way again → Support recommended.
2	Student	“I don't know how to ask for help” → Talk to a mentor → Evening → Request private support. The tracker shows “Request created”; the inbox confirms.
3	Support team — care@bput.ac.in / Care@2026	Page 10 → Support queue → Assign to me → Mark contacted → Schedule follow-up → Resolve with an outcome. The student's tracker and inbox follow every step.
4	Admin — control@bput.ac.in / Control@2026	Page 10 → Student Support Overview: totals only, small numbers hidden.

Limitations

- Emergency and help contacts must be configured by the institution before real use; until then students see generic guidance.
- One support-team role handles both counsellor and mentor requests; the student's preference is shown so the team can route it.
- Only the student's own voluntary check-ins are used as signals — attendance and other records are deliberately not used.
- Thresholds in supportRules.js are sensible defaults, not clinically validated; the institution's counselling service should review them.
- This system supports early help and human connection. It does not replace qualified professionals.